Do RL-Trained Memory Agents Forget Correctly?

Evaluating and Improving Conflict Resolution in Mem-α

Project type: Experimental evaluation and lightweight extension of an existing reinforcement learning-based memory system.

Problem and Motivation

LLM agents increasingly use external memory to retain information across long interactions. However, long-term memory must not only preserve information; it must also adapt when previously stored facts become outdated. This creates a continual learning challenge: incorporating new information without losing unrelated knowledge.

Mem-α (Wang et al., 2025) uses reinforcement learning to train a memory agent to organize information across core, semantic, and episodic memory through insert, update, and delete operations. Although Mem-α demonstrates strong performance on retrieval and long-context tasks, its evaluation excludes Conflict Resolution, a key dimension of MemoryAgentBench.

We investigate whether its learned memory-management policy can correctly replace outdated facts while preserving valid information.

Research Questions

1. **Conflict resolution:** How effectively does Mem-α incorporate new facts that contradict previously stored information, compared with untrained and prompted baselines?
2. **Memory behavior:** When an agent fails to resolve a conflict, is the failure caused by incorrect memory operations, retained outdated information, or downstream retrieval and reasoning?
3. **System improvement:** Can an explicit conflict-checking step before memory writes improve update accuracy without causing unnecessary forgetting?

Related Work

Our project builds on *Mem-α: Learning Memory Construction via Reinforcement Learning* (Wang et al., 2025), which introduces RL-trained memory management, and *Evaluating Memory in LLM Agents via Incremental Multi-Turn Interactions* (Hu et al., ICLR 2026), which introduces MemoryAgentBench and its FactConsolidation tasks. We

will also consider established retrieval-based and agentic memory baselines discussed in these works.

Approach

We will use the released Mem-α-4B checkpoint and existing implementation without retraining.

1. **Baseline evaluation:** Evaluate RL-trained Mem-α, base Qwen3-4B, and GPT-4.1-mini using the same memory architecture and downstream answering model. We will first verify the pipeline on a small published evaluation subset, then evaluate conflict resolution on FactConsolidation single-hop and multi-hop tasks, initially using 6K and 32K contexts.
2. **Memory-operation analysis:** Log memory states and tool calls to identify whether outdated information was correctly updated, retained as a current fact, duplicated, or incorrectly deleted. We will manually inspect approximately 50 selected update events, including successful and unsuccessful cases, and examine retention of unrelated facts.
3. **Conflict-aware memory updates:** Implement a lightweight retrieval-before-write policy that searches existing memory for potentially conflicting entries and explicitly checks whether a new fact should replace an existing fact or coexist with it. Compare this extension against unmodified Mem-α.

Evaluation Plan

Our primary metric is updated-fact question-answering accuracy, using the benchmark's scoring protocol and reporting single-hop and multi-hop results separately.

We introduce a diagnostic active stale-memory rate, defined as the proportion of superseded facts still represented as currently valid after an update.

We will additionally measure retention accuracy on non-conflicting facts, memory size in tokens, memory-operation patterns, and the inference overhead of conflict checking.

We hypothesize that RL-trained Mem-α will outperform its untrained counterpart but may still retain outdated facts. Explicit conflict checking may reduce these errors, although overly aggressive updates could remove valid information.

Milestones

Date	Milestone
Nov 18	Complete model setup, evaluation-pipeline verification, and initial baseline run
Dec 2	Complete the main conflict-resolution experiments, memory-state logging, and error analysis
Dec 17	Evaluate the conflict-aware extension, compare results, and complete the final report.

Expected Contributions

We aim to provide the first targeted analysis within this project of Mem-α's conflict-resolution behavior, distinguish failures of memory updating from failures of retrieval and reasoning, and evaluate a lightweight system modification that improves memory consistency without additional RL training.